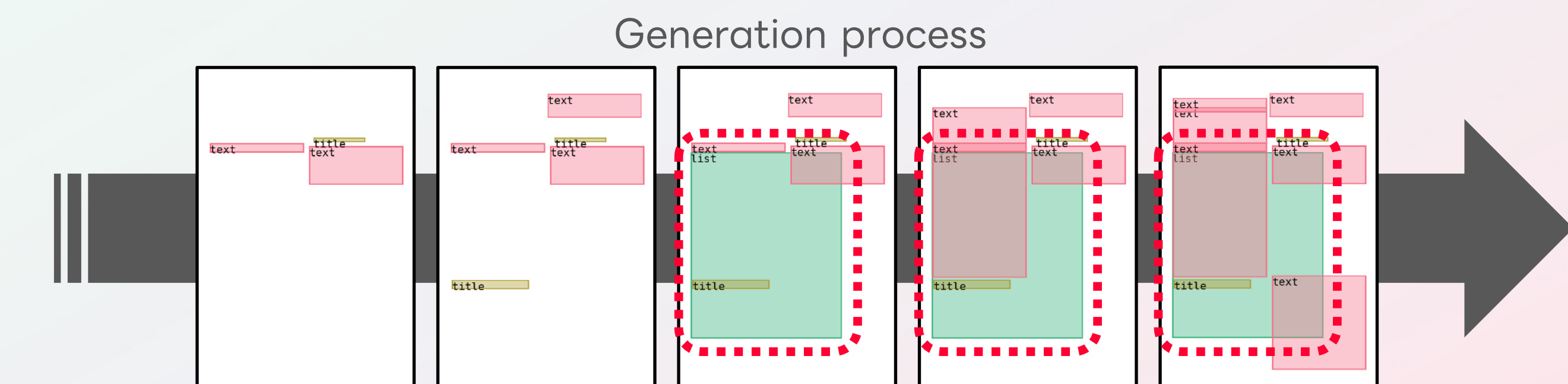


TL;DR

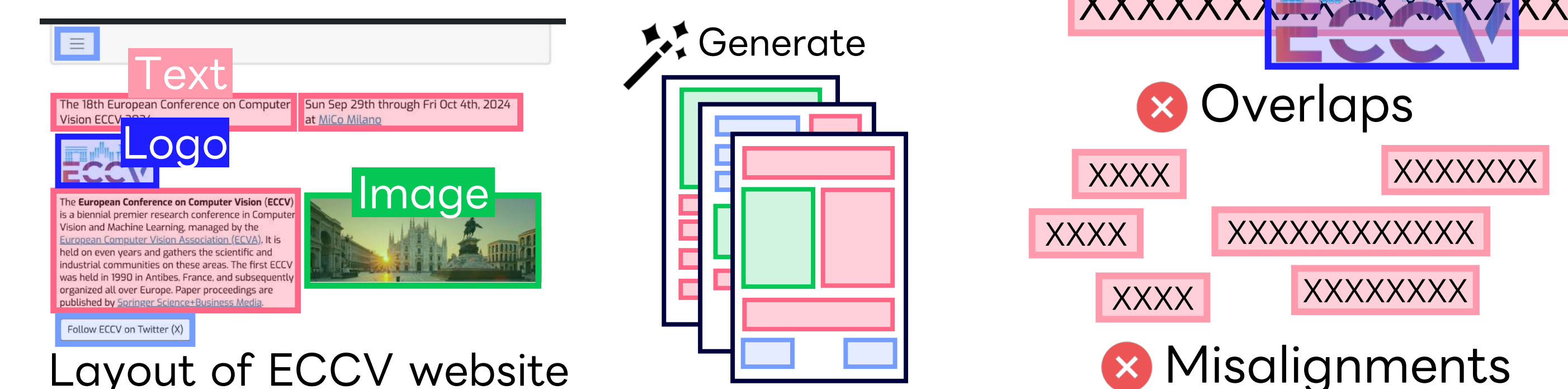
Existing Discrete Diffusion Model struggles to correct generation errors
= Generation errors got stuck! (**Layout-Sticking Problem**)



→ Our **Layout-Corrector** resets errors and prompts correction

Background: Layout Generation Goal: Correct Errors

Determine category, position, and size of each element (e.g., text, image)



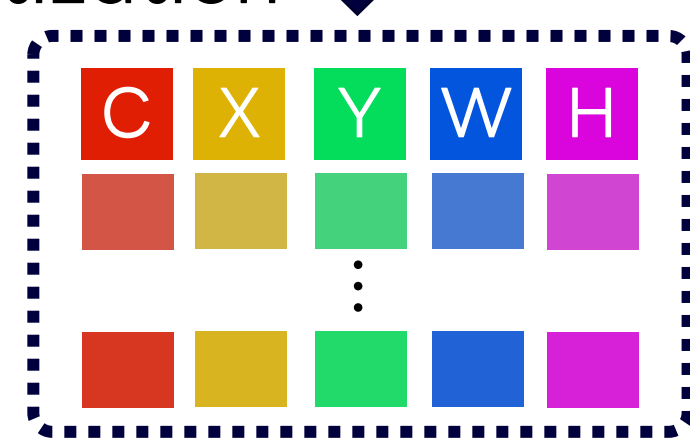
Preparation: Layout Generation using Discrete Diffusion Model (DDM)

[Inoue+, CVPR2023]

Layout as Discrete Tokens

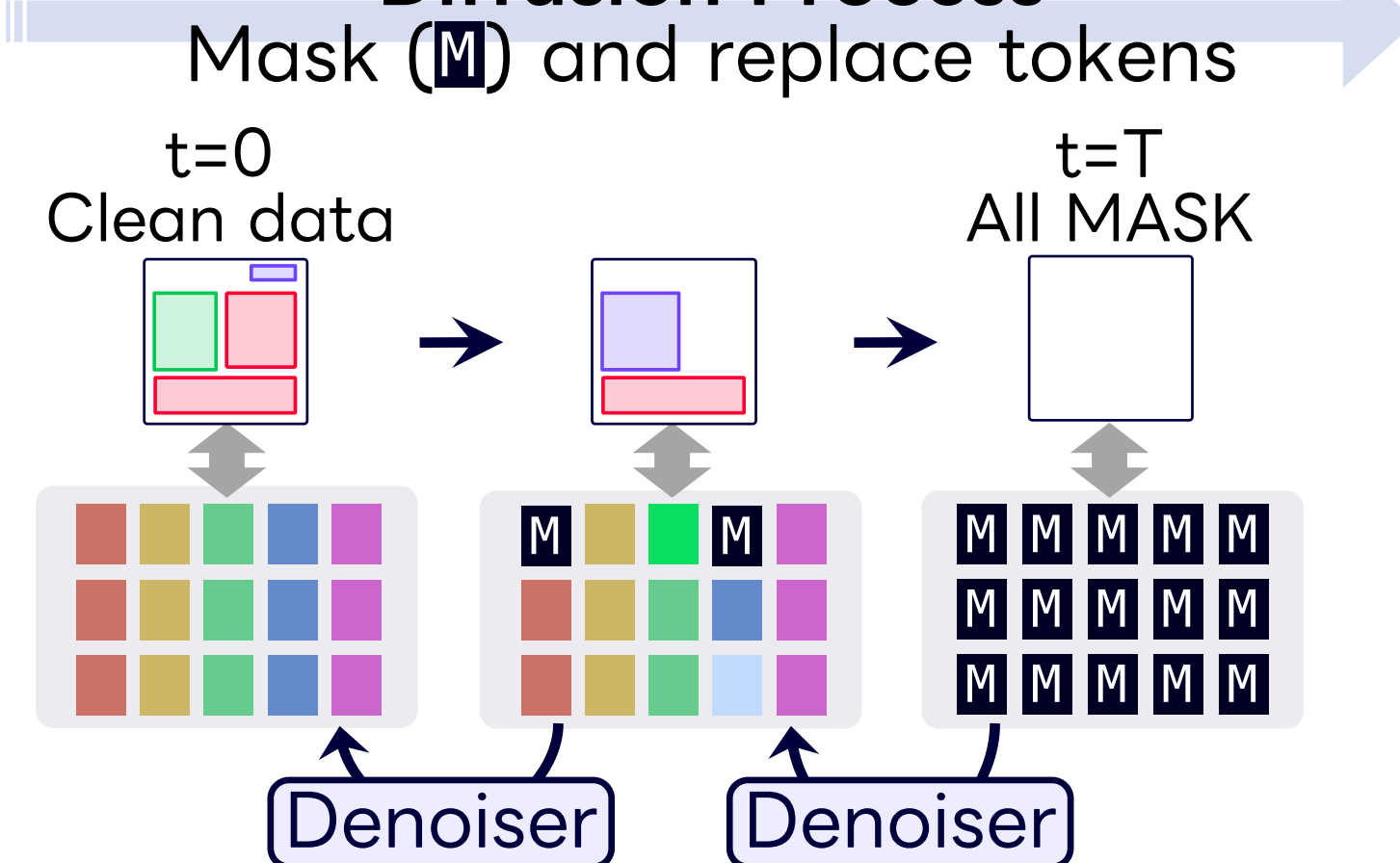
Category	X	Y	W	H
Text	15	10	100	50
...	...	...	...	...
Button	120	200	80	20

Quantization



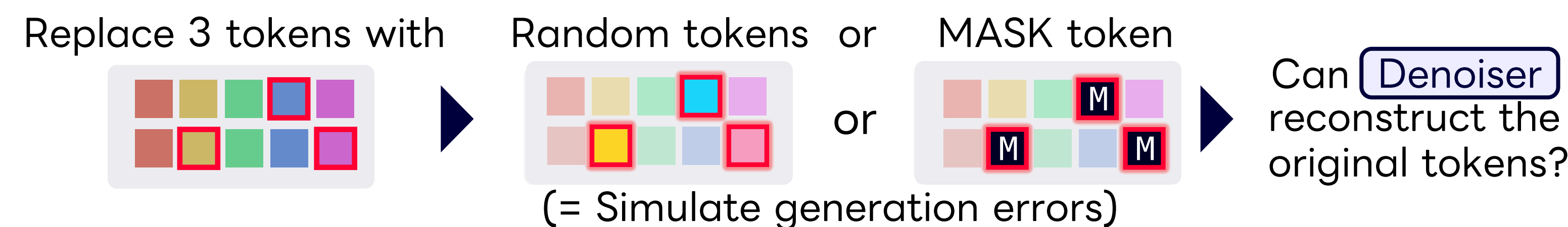
Layout token sequence

Diffusion Process



Backward Diffusion Process
Unmask and replace tokens

Preliminary: DDMs Struggle to Correct Error Directly



Reconstruction success rate

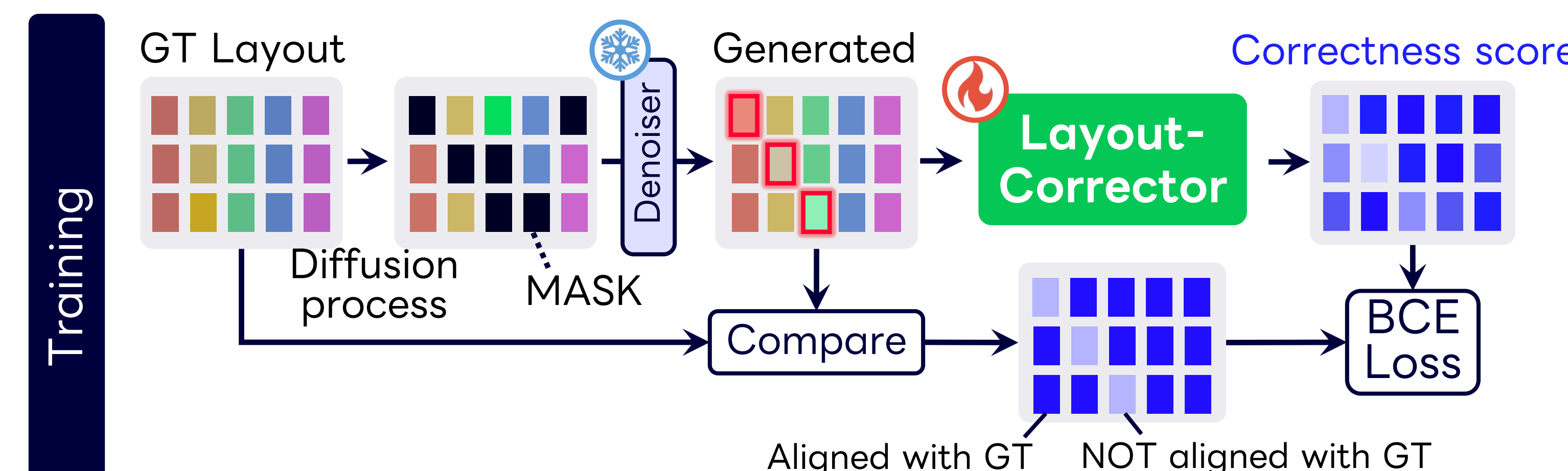
Replace tokens with:	$\beta = 0.0$	$\beta = 0.05$	$\beta = 0.1$
Random tokens	0.0%	13.2%	15.1%
MASK	35.4%	30.8%	29.7%

Hyperparameter β : Probability that replacement between non-MASK tokens happens during forward process

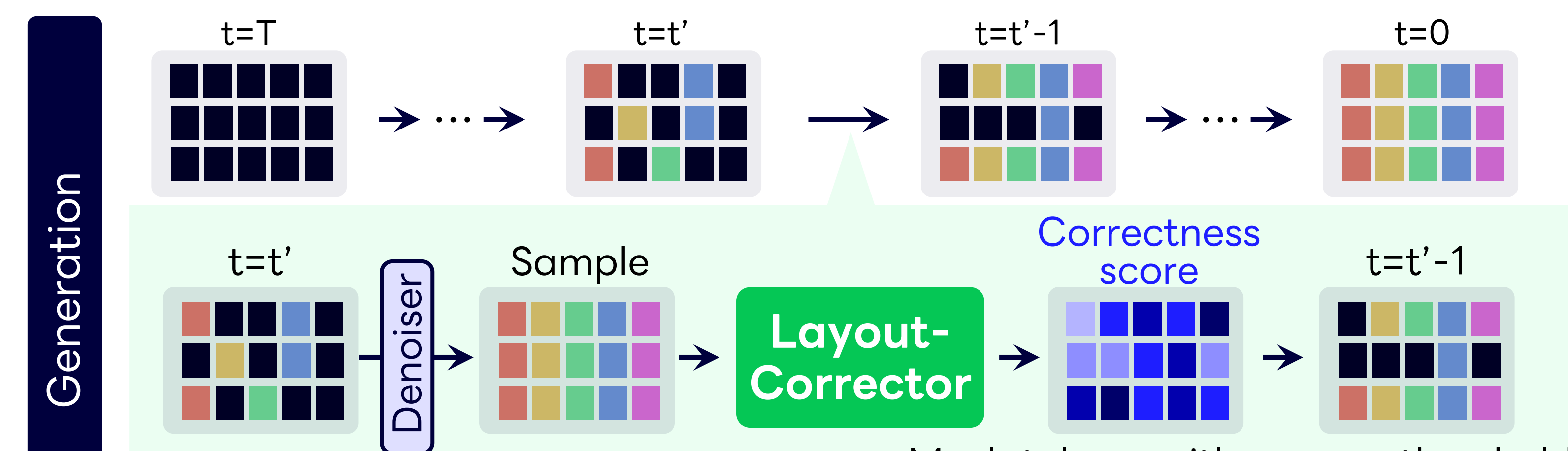
- ✗ DDMs struggle to correct erroneous tokens directly. (Success rate: ~15%)
- ✓ By masking erroneous tokens first, DDMs could correct them. (Success rate: ~35%)

Proposed Method: Layout-Corrector

Detects erroneous tokens to mask them → Prompt Correction



Predict whether each generated token aligns with the original (GT) token



Layout-Corrector is used at **only 3 timesteps** out of 100 to reduce overhead

Results

Layout-Corrector improves the performance of multiple baseline models

(Rico dataset)	FID ↓	Precision ↑	Recall ↑
MaskGIT	70.4	0.793	0.437
+Token-Critic	15.7	0.682	0.843
+Layout-Corrector	14.4	0.814	0.744
LayoutDM	6.37	0.759	0.906
+Token-Critic	18.0	0.884	0.670
+Layout-Corrector	4.79	0.811	0.891
VQ-Diffusion	7.83	0.716	0.907
+Token-Critic	15.2	0.842	0.731
+Layout-Corrector	5.29	0.809	0.898

Bold: improved compared to the baseline

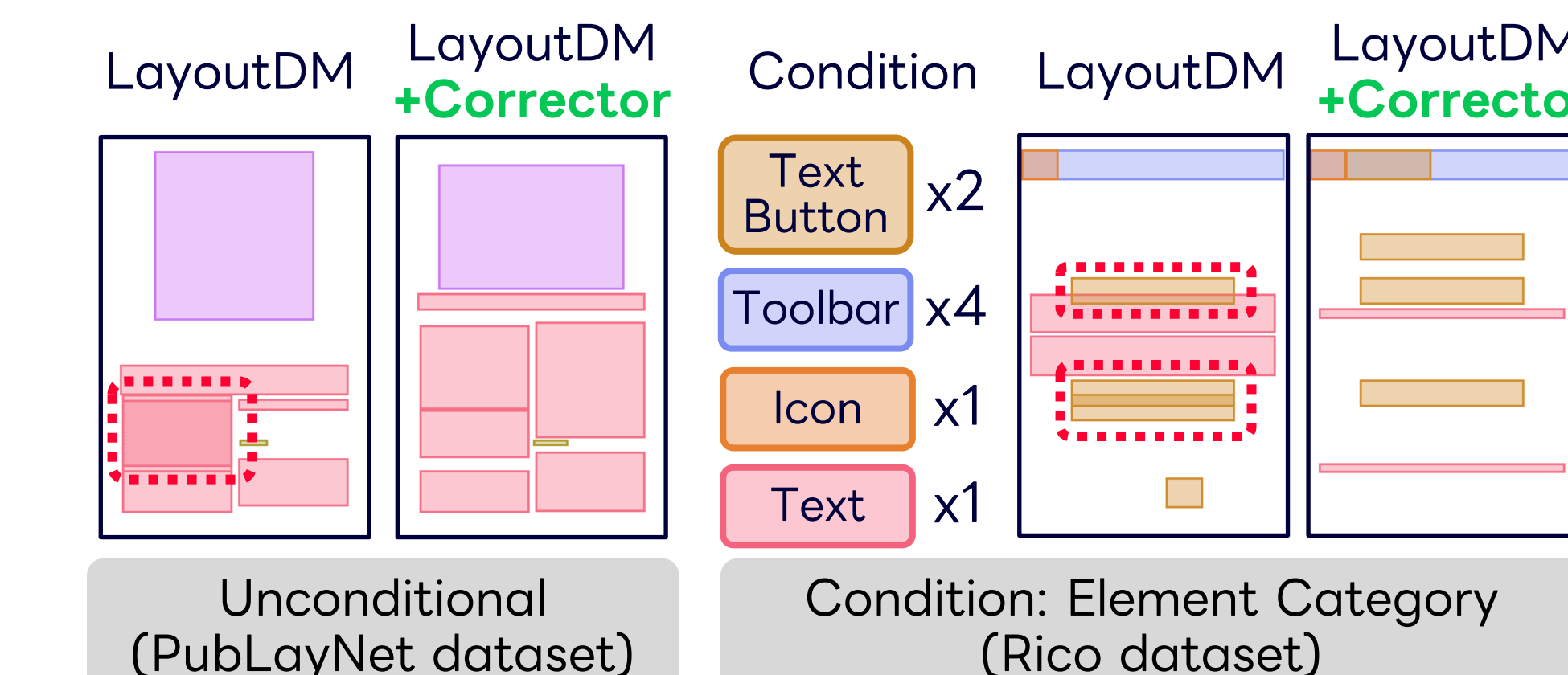
MaskGIT: [Chang+, CVPR2022]

VQ-Diffusion: [Gu+, CVPR2022]

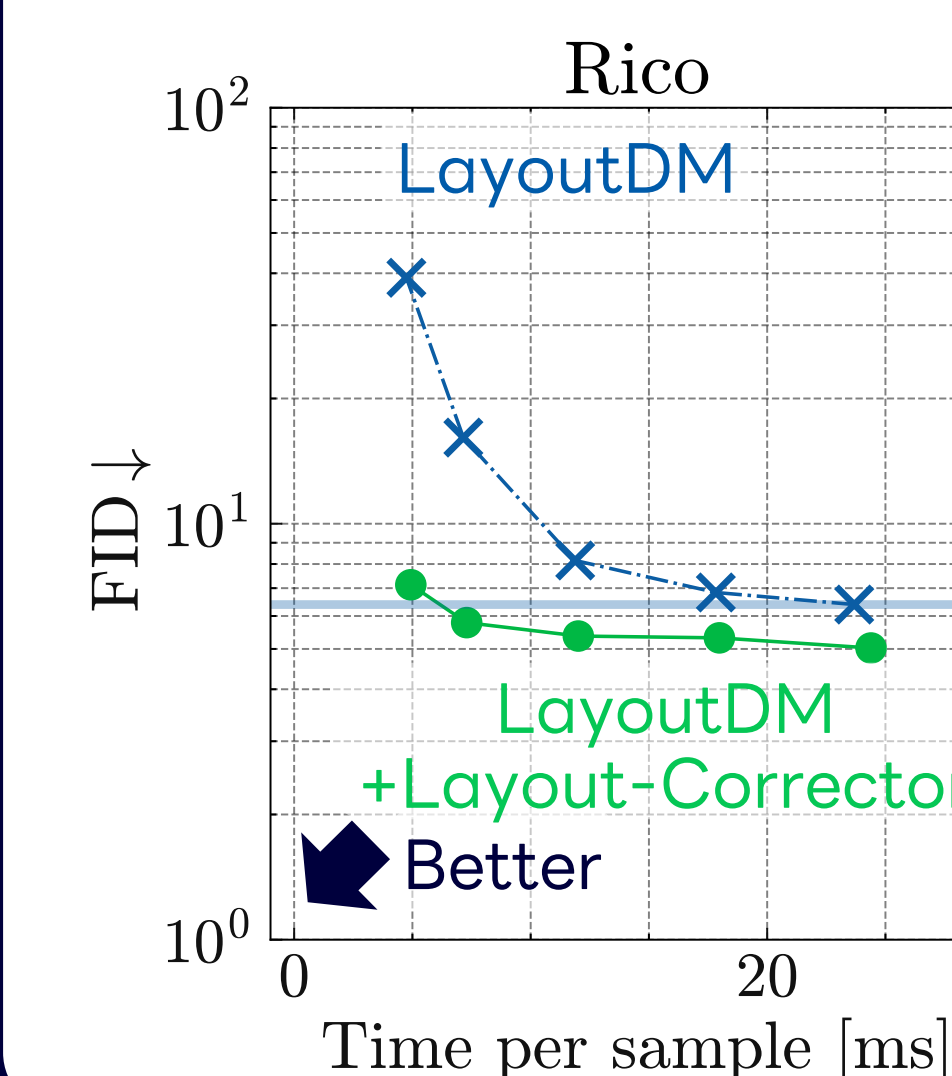
LayoutDM: [Inoue+, CVPR2023]

Token-Critic: [Lezama+, ECCV2022]

Visual samples: Errors are corrected



Speed-quality trade-off



Fidelity-diversity trade-off

